

Towards Trusted Federated Learning via Adaptive Trust-Flow in Edge Computing: A Study on Key Technologies

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Abstract

Federated learning has become a key enabler of edge intelligence by supporting collaborative model optimization while keeping data local. However, open edge-computing environments expose existing robust aggregation methods to difficult tradeoffs among security, robustness, and generalization, especially under non-independent and identically distributed (Non-IID) data, device heterogeneity, and untrusted clients. To address these issues, this paper proposes a trust-flow-driven trusted federated aggregation mechanism that integrates client admission, runtime monitoring, and secure aggregation from a hardware-software co-design perspective. Remote attestation and runtime monitoring based on trusted execution environments (TEEs) are first used to enforce client admission and verify execution integrity during local training. To identify stealthy cross-round attacks, we then design an orthogonal dual-stream reputation evolution method that decomposes client trust states into a historical utility stream for long-term contribution and a short-term risk stream for sudden anomalous behavior. These dynamic trust signals guide a layer-wise differentiated aggregation strategy based on a clean reference direction, where risk gating, weight re-normalization, and dynamic clipping suppress malicious updates while preserving legitimate heterogeneous contributions. Experiments on Flower with CIFAR-10, 20 clients, and six mixed attack types show that the proposed mechanism achieves an average accuracy of 92.31% ($\pm 0.09\%$) while limiting the attack success rate (ASR) to 10.21% ($\pm 0.05\%$), with the permanent false positive rate on benign clients approaching zero.

Keywords: Federated learning, Edge computing, Trusted execution environment, Risk state modeling, Layered robust aggregation, Backdoor defense

1. Introduction

Current advancements in terminal device computing power enable efficient local model training using on-device data. Meanwhile, to better aggregate multi-device training experiences for collaborative model optimization while preserving local data privacy, the federated learning paradigm has been extensively studied [1]. Its core feature—exchanging only model parameters without offloading data—is widely applied in distributed collaborative training, serving as a critical pillar for edge intelligence. However, as the interconnected network environment becomes increasingly open, the performance bottlenecks of federated learning no longer reside solely in model architecture or optimizer design. The trustworthiness and robustness of the aggregation phase have emerged as vital factors determining overall availability. In practical scenarios, edge devices often exhibit significant heterogeneity in both data distribution and processing capabilities. Furthermore, fully constraining the behaviors of participating clients remains challenging. Such characteristics inevitably amplify statistical drifts and introduce security risks during empirical aggregation.

In standard federated learning paradigms, clients upload parameters after completing local data processing and model

training. The server then aggregates these local updates and distributes the updated global model, iterating this process until convergence. Evidently, this workflow encounters multiple vulnerabilities within the increasingly open, multi-edge interconnected network. Open training environments inherently lack mandatory integrity constraints. Malicious clients can bypass actual training, directly crafting and uploading poisoned parameters, thereby severely impairing the server's ability to verify the authenticity of aggregated updates. Several studies propose methods based on trusted execution or verifiable training [2, 3, 4], employing hardware-assisted remote attestation to ensure local training authenticity. Nevertheless, these mechanisms fail to address scenarios where the training process is genuine but the model updates are injected with malicious features. Furthermore, attackers execute parameter poisoning or stealthy backdoor attacks [5, 6] by embedding malicious triggers within normal update distributions, drastically reducing the success rate of single-round detection. To alleviate the instability of single-round assessments, researchers have introduced methods based on historical reputation and similarity [7, 8] that leverage cross-round information. Yet, these approaches suffer from delayed recognition of sleeper attacks, where adversaries accumulate reputation early on before engaging in persistent, low-amplitude poisoning. Additionally, single-reputation scores combined with fixed-threshold mechanisms remain highly inadequate for identifying deep abnormal

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trigger features.

The presence of non-independent and identically distributed (Non-IID) data across devices increases the directional dispersion of optimal updates. This heterogeneity easily blurs the boundary between normal statistical fluctuations and anomalous deviations, creating a conflict between robustness and generalization under strong Non-IID conditions. Certain approaches rely on distance-based exclusion or statistical truncation to suppress abnormal updates [9, 10], while others adaptively adjust weights to mitigate heterogeneous impacts [11]. Under severe long-tail distributions, however, such methods often misclassify legitimately drifted parameter updates as anomalies. This misjudgment can lead to the excessive exclusion of crucial heterogeneous information, significantly degrading the model’s generalization capability. In real-world edge intelligence scenarios, these challenges—open execution environments, stealthy cross-round attacks, and pronounced heterogeneous statistical drifts—rarely occur in isolation. Multiple factors frequently compound, leading to intractable dilemmas such as elevated false positive rates from tightening interception thresholds or increased false negative rates when relaxing them. Consequently, establishing a unified state modeling and coordinated decision-making mechanism spanning the entire workflow remains a critical challenge. Such a mechanism must balance client admission and trustworthiness during local training with low false positive and high recall requirements during aggregation, all while maintaining the convergence and generalization performance of collaborative model optimization.

To tackle the three major challenges outlined above, this paper proposes a trusted federated aggregation mechanism driven by a full-process trust flow from a hardware-software co-design perspective. This mechanism follows the execution pipeline of federated learning. During client admission and local training, TEE remote attestation, runtime feature entropy analysis, and Kalman filtering are introduced to formulate a dynamic admission trust score that updates across rounds. At the server auditing stage, the current-round content quality is evaluated based on a clean reference direction. The client’s state is then decoupled into a historical utility flow and a short-term risk flow, using Bayesian posterior inference and risk EMA to characterize long-term contributions and short-term anomalies separately. During aggregation, layered risk gating, survivor weight re-normalization, and dynamic clipping are applied based on these trust states. This strategy suppresses malicious updates while retaining legitimate heterogeneous contributions, mitigating the tradeoff between security and generalization under Non-IID conditions.

The primary contributions of this paper are:

1. **Proposing a TEE-anchored dynamic admission mechanism for edge clients.** Remote attestation defines the client admission boundary. Variations in gradients, losses, and instruction distributions during training are then transformed into entropy features. By leveraging Kalman filtering to update the *TrustScore*, admission control extends from a one-time static verification to a continuously evolving trust metric.
2. **Developing a dual-stream orthogonal reputation evolution mechanism against stealthy attacks.** Client states are decoupled into a historical utility stream (HistPerf) and a short-term risk stream (RiskEMA), independently handling long-term contributions and sudden anomalies. A *RawScore* connects content auditing, state transitions, and subsequent aggregation control, balancing malicious recall with the suppression of permanent false positives.
3. **Designing a layered risk gating aggregation strategy tailored for strong Non-IID scenarios.** Guided by a server-side clean reference direction, layer-wise security sensitivities, and dynamic L2 clipping, this approach applies differentiated admission thresholds and survivor weight re-normalization across network layers. It reduces the risk of deep backdoor penetration while preserving valid updates from legitimate long-tail nodes.
4. **Implementing and validating the framework end-to-end under mixed attack scenarios.** Experiments involving 20 clients and 6 mixed attack types ($n = 5$) were conducted on the Flower platform. Results demonstrate that under high-adversarial settings, the proposed framework maintains an accuracy of 92.31%(±0.09%) and restricts the ASR to 10.21%(±0.05%), with the permanent false positive rate approaching zero.

2. Background and Related Work

2.1. Federated Learning and Edge Computing Collaborative Architecture

Federated learning (FL) completes global model training through parameter-level collaboration under the constraint that data remain local. In edge computing scenarios, the client-edge-cloud architecture has been widely adopted: the edge side performs local training and uploads updates, while the central server handles aggregation and model distribution.

In the standard FL setting, the global optimization objective is to minimize the empirical risk over the data distributions of all clients:

$$\min_W \mathcal{L}(W) = \sum_{k=1}^K p_k \mathcal{L}_k(W) \quad (1)$$

where $p_k = |\mathcal{D}_k| / \sum |\mathcal{D}_j|$ denotes the data-volume proportion. In round t , the server distributes the global model $W_{global}^{(t-1)}$. Client k takes it as the initial model and runs SGD for E local epochs with learning rate η_l , yielding the local update:

$$\begin{aligned} \Delta W_k^{(t)} &= W_{k,E}^{(t)} - W_{global}^{(t-1)}, \\ W_{k,e}^{(t)} &= W_{k,e-1}^{(t)} - \eta_l \nabla \mathbb{E}[\ell(W_{k,e-1}^{(t)}; x, y)]. \end{aligned} \quad (2)$$

Here, the expectation $\mathbb{E}[\cdot]$ denotes mini-batch sampling over the local dataset $\mathcal{D}_k$. The client then uploads its update, and the server applies a form of parameter aggregation: $W_{global}^{(t)} = W_{global}^{(t-1)} + \text{Agg}(\{\Delta W_k^{(t)}\}_{k \in \Phi})$.

This architecture, however, naturally faces communication latency, device heterogeneity, and Non-IID data. To mitigate

these issues, prior work has proposed methods such as Clustered FL [12], adaptive pruning-expansion (FedPE) [13], and split federated learning (ParallelSFL) [14]. As system complexity increases, the attack surface also expands, giving malicious nodes more room to remain hidden and interfere with training.

2.2. Edge Impersonation, Model Poisoning, and Stealthy Backdoor Attacks

In open federated networks without mandatory admission control, the server has difficulty accurately judging client identities and update quality. Attacks mainly fall into two categories. The first is **untargeted poisoning**, such as Sign-flipping and Gradient Scaling, which aims to disrupt global convergence and cause denial of service. The second is **targeted backdoor attacks**, including label flipping, clean-label poisoning, and semantic backdoors, which induce targeted misclassification at inference time through concealed triggers.

2.3. Mechanisms and Inherent Limitations of Existing Active Defenses

To address these threats, early defenses largely relied on geometric and statistical filtering, such as Krum[9] and Trimmed Mean[10]. More recent studies have introduced temporal trust evaluation, including FLTrust[7], FoolsGold[8], and RaSA[11], while continuing to expand into malicious detection[15], trusted training frameworks[2], hierarchical protection[16], and related directions. For backdoor and collusion attacks, schemes such as FLPurifier[17], RoseAgg[18], and ShieldFL[19] have also been developed. Purely software-based defenses still have clear weaknesses under strong Non-IID conditions and high-privilege attacks. Tightening thresholds raises the false positive rate (FPR), while relaxing them may allow dormant attacks to pass. When attackers control the underlying environment, software probes that lack a hardware root of trust also struggle to form a complete defensive loop.

2.4. Trusted Hardware Root of Trust in Federated Environments (TEE)

To fill the gap in physical trust, research has begun to introduce trusted execution environments (TEEs) such as Intel SGX and ARM TrustZone. Through isolated execution and protected memory, TEEs preserve the integrity of critical code and key material, making it difficult for an attacker to directly tamper with the attestation process even after the host operating system has been compromised[3, 20]. Existing work shows that TEEs have practical value in training integrity[4], threat mitigation[21], and trust sharing in industrial IoT[22]. Their federated extensions and efficiency optimization in SGX environments have also been validated[23, 24]. On this basis, this paper builds a hardware-software collaborative defense system. Unlike methods that only perform static encryption or single-dimensional anomaly evaluation, this work temporally couples hardware remote attestation with dual-stream reputation evolution. The cloud side is responsible for parameter auditing, while the edge-side TMAA provides trusted evidence and runtime monitoring, forming a coordinated protection mechanism that combines mandatory admission with dynamic circuit breaking.

3. System Model and Problem Definition

3.1. Client-Edge-Cloud Collaborative Architecture and System Workflow Modeling

This paper considers a typical client-edge-cloud heterogeneous communication environment (see Figure 1). The system consists of a central **aggregation server (Server, S)** and K edge clients, denoted by $C = \{c_1, c_2, \dots, c_K\}$. Clients may join or leave dynamically, and some nodes may be controlled by attackers.

Based on the network interaction topology above, the proposed trust-flow-driven trusted federated framework builds a complete trust transmission and evaluation chain between the edge and the cloud. The overall workflow of the system and the relationships among its main physical entity modules are defined as follows:

1. **Edge client side (trusted execution and feature extraction)**: Each admitted client must be equipped with a trusted execution environment (TEE) and a trusted management agent (TMAA) deployed inside it. During local training, the TMAA generates an Attestation Quote before training to statically verify the integrity of the execution environment. It also continuously monitors key behavioral feature sequences during training, such as changes in gradient direction and loss fluctuation. Information entropy and Kalman filtering are then used for an initial dynamic trust assessment, which is uploaded to the cloud together with the model update parameters.
2. **Central server side (auditing, evolution, and aggregation control)**: As the global coordinator, the server receives client updates and trust evidence, then performs a three-stage coordinated defense based on trust flow. It first conducts **content auditing**, extracting a clean reference direction and using cosine similarity to calculate high-dimensional update quality. It then enters **dual-stream state evolution**, where Beta Bayesian inference is used to build a historical utility stream for long-term contribution evaluation, while multidimensional max-value probes update a short-term risk stream to identify sudden anomalies. Finally, **layered robust aggregation** applies differentiated gated clipping and survivor weight re-normalization according to the security sensitivity of different network layers, producing and distributing a globally secure update.

Through the coordinated interaction of the edge and cloud entity modules, the system forms a full-lifecycle physical and logical closed loop: hardware admission control $\rightarrow$ runtime feature monitoring $\rightarrow$ trust state evolution $\rightarrow$ risk-gated aggregation.

3.2. Composite Threat Assumptions and Strong Trust-Base Boundary (Threat Model)

To clearly define the boundary between attacker and defender capabilities, this paper adopts the following threat model (see Figure 2):

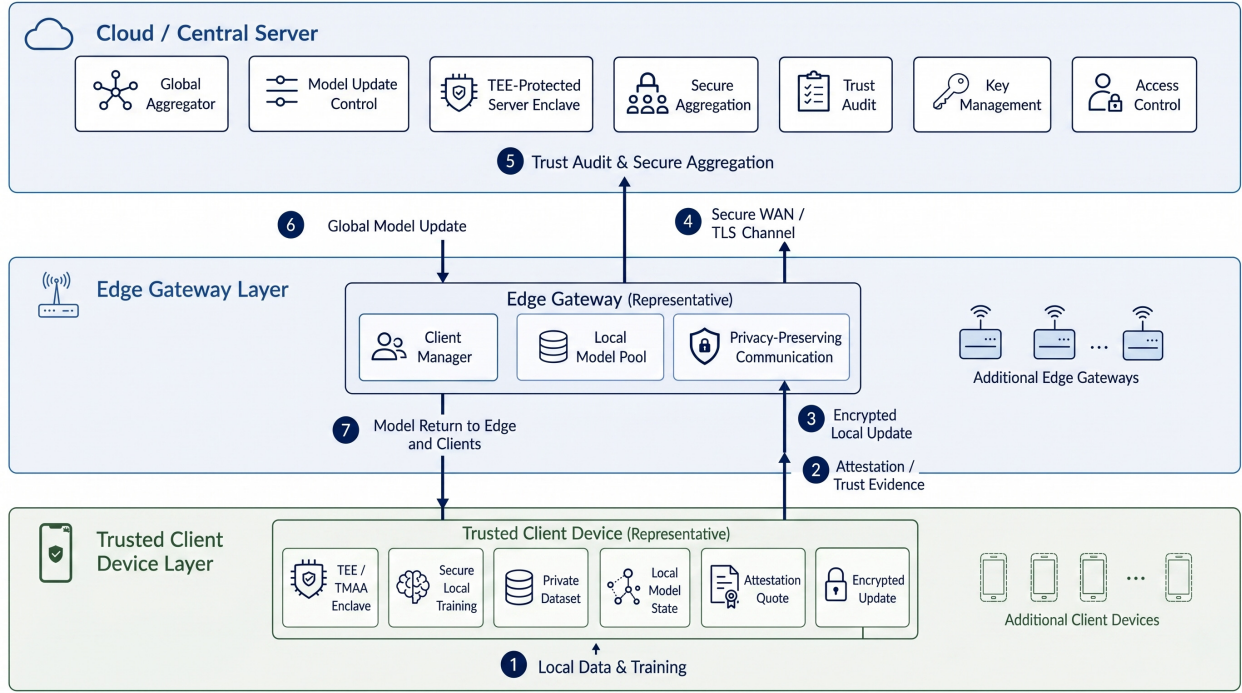


Figure 1: Federated computing network interaction: client-edge-cloud collaborative physical architecture based on TEE-anchored encryption

- **Hardware isolation boundary (root of trust):** Under standard security assumptions, excluding advanced attacks such as microarchitectural side channels and physical probing[3], TEEs such as TrustZone/SGX can preserve the integrity of critical execution and attestation processes even if the host operating system is compromised.
- **Honest-but-curious server:** The server follows the protocol for dual-stream scoring and layered aggregation, but it may attempt to infer additional private information about clients.
- **Internal malicious clients (Internal Attackers/Poisoners):** Attackers may possess valid admission identities and locally perform label tampering, sample poisoning, sign flipping, gradient scaling, and related operations. This paper assumes an upper bound of $< 50\%$ malicious clients. The main experiments use 30% as a typical high-risk setting and additionally report a 50% boundary stress test.

3.3. Core Defense Optimization Objectives of the Security Architecture (Design Goals)

For highly heterogeneous data and a high proportion of malicious participants, the design goals of the proposed framework are as follows:

1. **High defense effectiveness and robust convergence:** Under multiple types of poisoning and backdoor attacks, the system must accurately identify malicious nodes, namely achieving high TPR and low ASR, while maintaining high availability and stable convergence of the global main-task model.

2. **High tolerance for heterogeneous long-tail nodes:** The framework should effectively overcome feature deviation caused by Non-IID data distributions, avoid mistakenly removing honest nodes that carry long-tail data, and keep the false positive rate (FPR) close to zero.
3. **Lightweight overhead and edge adaptability:** Without introducing the computational burden of full-scale cryptography, such as fully homomorphic encryption, the framework should build an efficient trusted protection mechanism whose computation and communication overheads remain compatible with the resource bottlenecks of heterogeneous edge devices.

4. Proposed Method: Trust-Flow Framework

For the threat model above, this paper proposes a trust-flow-driven trusted federated learning framework (see Figure 3). Following the execution order of federated learning, the framework first performs trusted admission and runtime monitoring on the client side, then conducts current-round content auditing and cross-round reputation evolution on the server side, and finally enters layered security control during aggregation. Each training round is therefore divided into five phases: pre-training admission, content auditing, state evolution, layered aggregation, and global distribution. At a higher level of abstraction, these phases correspond to three layers: client admission and monitoring, server auditing and evolution, and aggregation security assurance.

The key of this macro-to-micro structure is the continuous transfer of trust states. Phase one outputs *TrustScore*, indicating whether the client has a trustworthy basis for admis-

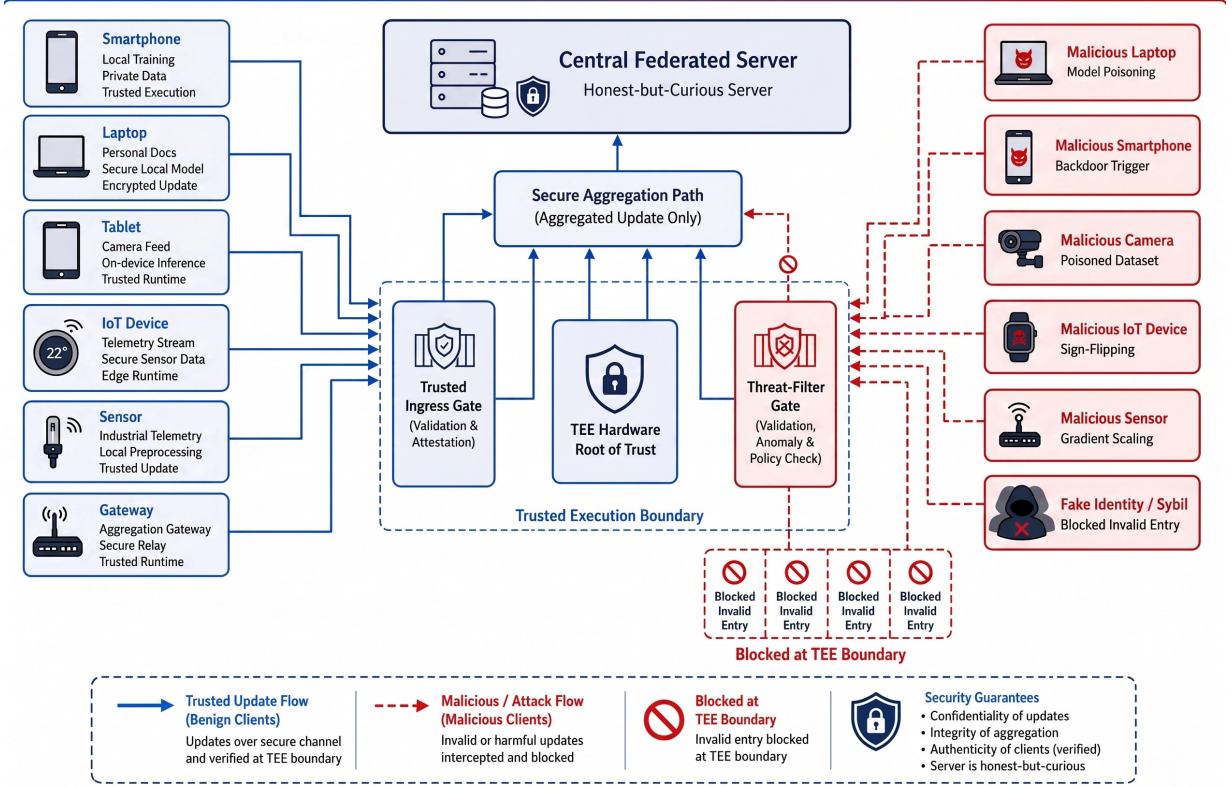


Figure 2: Security penetration and multidimensional composite poisoning threat model in an edge wide-area network

sion and execution. Phase two outputs *ContentScore*, judging whether the update submitted in the current round deviates from the clean reference and the collaborative direction of the group. Phase three deposits current-round signals into *HistPerf* and *RiskEMA*, distinguishing long-term low contribution from short-term high risk. Phase four fuses the three types of signals into *RawScore* and performs differentiated aggregation by network layer. Unlike single-point defenses, data generated at each phase continues to constrain later phases, continuously updating the **Trust Flow** and forming closed-loop control.

To facilitate the following derivation, Table 1 lists the definitions of core symbols.

4.1. Hardware Admission and Dynamic Trust Assessment

Client admission should not be limited to one-time identity verification. It requires a tracking mechanism that reflects the continuing trustworthiness of local training. For this purpose, the initial trust level is decoupled into a two-layer architecture of static admission and dynamic evaluation. The former relies on underlying hardware to judge the legitimacy of the execution environment, while the latter uses the Information Theoretic Learning (ITL) paradigm[25] to convert transient behavioral fluctuations during training into dynamic observation signals, producing a transferable *TrustScore* through filtering.

1. **Static Hardware Admission:** During the initialization of federated computing, each participating node must invoke

its internally isolated device-unique key (DUK) and initiate a remote attestation quote to the central server, including the chain values of platform configuration registers (PCRs). The server verifies memory code segments and runtime signatures, then grants initial communication privileges to nodes that pass verification ($M_{attest,k} = 1$).

2. **Feature Sequence Extraction and Entropy Analysis:** In anti-poisoning trust assessment for distributed systems[26], a single static measurement is insufficient to capture complex runtime mutations. The client-side TEE is therefore configured to continuously monitor key feature sequences during the training cycle, $X_k = \{\Delta grad, \Delta loss, instr_dist\}$. The information entropy $H(X_k)$ of this sequence is quantified as:

$$H(X_k) = - \sum p(x_i) \log p(x_i) \quad (3)$$

A high entropy value in the sequence usually indicates greater uncertainty in system operation. In physical terms, this often corresponds to backdoor-controlled execution, severe disturbance in gradient direction, or abnormal underlying instruction distributions.

3. **ITL-Driven Dynamic Trust Evolution (Kalman-based):** To prevent abrupt anomalies from contaminating the trust state, this framework constructs an entropy-driven adaptive covariance mapping mechanism. Instead of assuming a constant prior, the system maps the dynamically captured information entropy to the measurement-noise

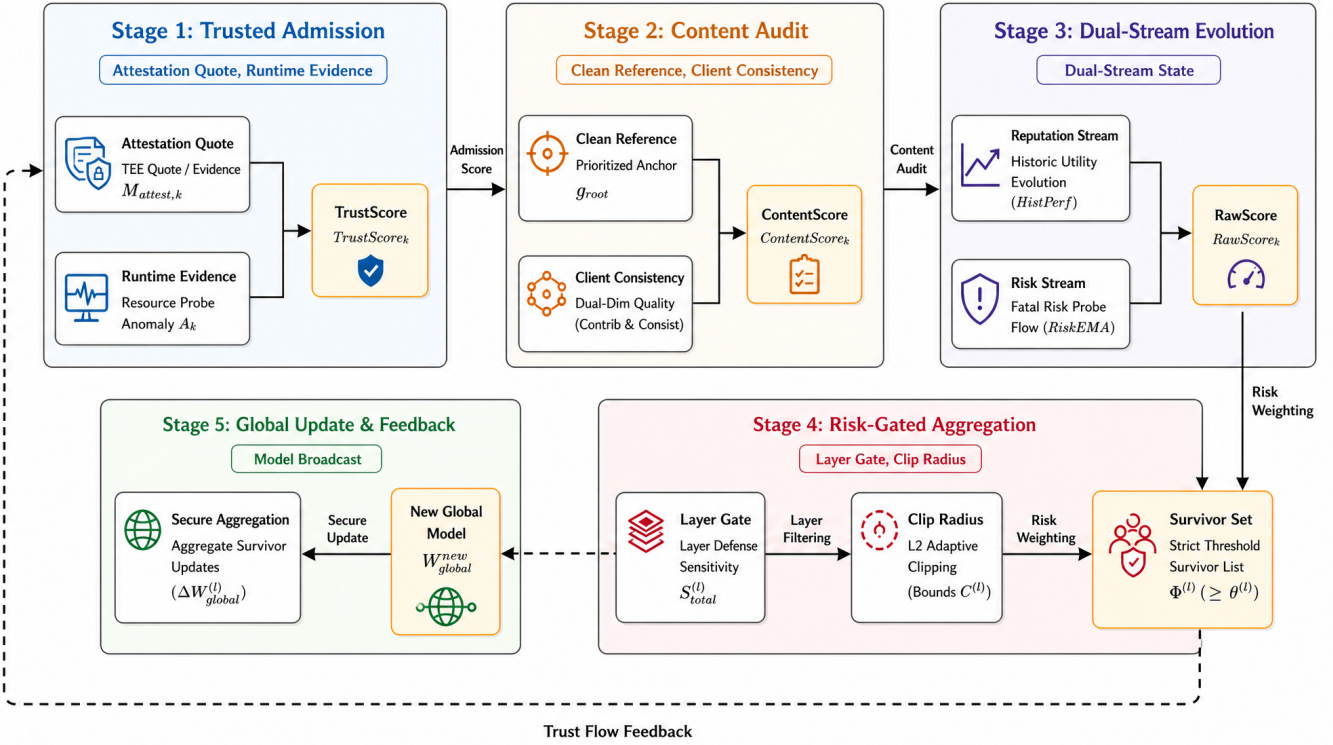


Figure 3: Overview of the five-phase defense closed loop in trust-flow-driven trusted federated learning

covariance matrix $R_k = f(H(X_k))$ in the Kalman observation equation. In the classical predict-update recursion:

- **Prediction stage:** Based on the posterior distribution from the previous round, the prior trust expectation $\hat{T}_k^-$ at the current moment is extrapolated.
- **Update stage:** The instantaneous behavioral features of the current round are introduced for measurement update, yielding the posterior trust estimate $\hat{T}_k$ and the corresponding state error covariance matrix P_k .

On this basis, combined with the high-entropy penalty property of anomalous sequences, the system introduces the following core state evolution rule:

$$\hat{T}_k^{(t)} = \hat{T}_k^{(t-1)} + K_k^{(t)}(Z_k^{(t)} - \hat{T}_k^{(t-1)}) \quad (4)$$

$$R_k^{(t)} = \exp(\lambda \cdot H(X_k^{(t)})) \quad (5)$$

In these equations, $Z_k^{(t)}$ maps the node's real-time baseline observed utility. When client behavior undergoes a high-risk jump, namely high entropy, the exponentially amplified $R_k^{(t)}$ rapidly reduces the Kalman gain $K_k^{(t)}$. This mechanism makes the filter automatically suppress the absorption of high-risk anomalous instantaneous observations, thereby maintaining the stable resilience of the posterior trust estimate.

Furthermore, by the mathematical nature of Bayesian filtering, the covariance matrix P_k iteratively produced by the filter objectively characterizes the confidence boundary of the node's dynamic trust estimate. A smaller P_k indicates that the current filter has more effectively suppressed measurement

noise and that the reputation score is more reliable. Phase one therefore does not output a coarse binary decision. Instead, it jointly evolves the expected mean and uncertainty produced by Kalman filtering, forming a temporally resilient admission privilege factor $TrustScore_k = \hat{T}_k^{(t)} \cdot (1/(1 + P_k^{(t)}))$.

4.2. Reference Direction Construction and Consistency Auditing

Relying only on cosine similarity among clients can be misled by collusion attacks. To address this issue, this paper introduces **Vanilla Reference Prioritization**. When the server holds a small isolated validation set (Pure Clean Set), it first computes g_{root_clean} as the reference direction. If this direction is unavailable, a fallback reference g_{ref} is constructed from updates submitted by high-trust, low-risk clients. The corresponding weight is defined as:

$$\omega_k^{ref} = TrustScore_k \cdot (1 - RiskEMA_k^{prev})^2. \quad (6)$$

The final reference direction g_{root} is:

$$g_{root} = \begin{cases} g_{root_clean}, & \text{if server validation direction available,} \\ \frac{\sum_k \omega_k^{ref} \Delta W_k}{\sum_j \omega_j^{ref}}, & \text{otherwise.} \end{cases} \quad (7)$$

After obtaining g_{root} , the server calculates a content quality score for each client update ΔW_k . This score considers both global contribution, namely consistency with the reference direction, and group consistency, namely the degree of agreement

Table 1: Core system parameters and trust-flow notation

Symbol	Definition	Symbol	Definition
$W_{global}^{(t)}$	Global model weights in round t	$\mathcal{A}$	Active node set eligible for valid aggregation in the current round
ΔW_k	Model gradient/parameter update submitted by client k	$\Phi^{(l)}$	Valid secure survivor subset at layer l
$TrustScore_k$	TEE hardware-anchored admission trust score	$HistPerf_k$	Accumulated state of the historical utility stream ($HistPerf \in [0, 1]$)
$ContentScore_k$	Current-round content quality score under dual references	$RiskEMA_k$	Decayed state of the short-term risk stream ($RiskEMA \in [0, 1]$)
$RawScore_k$	Final privilege score after fusing three-dimensional scores and applying risk discounting	$M_{attest,k}$	Hardware remote attestation verification flag
g_{root}	Clean guiding anchor direction extracted by the server	A_k	Runtime anomaly score of client k in physical resource usage

with other high-reputation clients:

$$S_{contrib,k} = \max(0, \alpha_1 + \alpha_2 \cdot \cos(\Delta W_k, g_{root})), \quad (8)$$

$$S_{consist,k} = \frac{\sum_{j \neq k} TrustScore_j \cdot \xi_{j,k}}{\sum_{j \neq k} TrustScore_j + \varepsilon}, \quad (9)$$

$$ContentScore_k = \frac{(1 + \beta_{fusion}^2) \cdot S_{consist,k} \cdot S_{contrib,k}}{\beta_{fusion}^2 \cdot S_{consist,k} + S_{contrib,k} + \varepsilon}. \quad (10)$$

Here, $\xi_{j,k} = \max(0, \alpha_1 + \alpha_2 \cos(\Delta W_k, \Delta W_j))$, and $\alpha_1 + \alpha_2 = 1.0$ adjusts the cosine mapping interval. The parameter $\beta_{fusion} > 1.0$, with default $\beta_{fusion} = 2.0$, moderately increases the weight assigned to consistency with the clean direction, strengthening resistance to collusive drift. $ContentScore_k$ describes only the content quality of the current-round update and is not equivalent to permanent reputation. For legitimate long-tail nodes under strong Non-IID settings, a low single-round score may result from data skew rather than attack behavior. This score is therefore passed to the cross-round state evolution in phase three, where subsequent handling is jointly determined by the historical utility stream and the short-term risk stream.

Implementation summary. In each round, the server first rejects clients that fail TEE attestation, updates $TrustScore_k$ from runtime entropy and Kalman confidence, constructs g_{root} from either the clean validation direction or high-trust low-risk clients, and then computes $ContentScore_k$ from reference and peer consistency.

4.3. Dual-Stream Evolution of Historical Utility and Short-Term Risk

In edge settings with strong non-independent and identically distributed (Non-IID) data, a simple “low score means elimination” rule can confuse two different cases: a benign client whose update is weak because of long-tail data, and a

malicious client whose update is risky because of poisoning. Treating both cases with one score raises the false positive rate (FPR). To separate them, this study models client state through two temporal streams that are updated independently and fused only at the decision stage: a **historical utility stream** ($HistPerf$) for long-term contribution and a **short-term risk stream** ($RiskEMA$) for sudden anomalies. The former asks whether a client has provided useful updates over time, whereas the latter asks whether the current behavior is unsafe. Keeping these signals separate reduces the conflict between false positives and false negatives in single-score reputation systems.

(1) Historical Utility Stream (Based on Bayesian Posterior Inference) This stream evaluates a node’s sustained contribution. We use a Beta reputation model, where the probability that client k provides a high-quality update is represented as $\text{Beta}(\alpha_k, \beta_k)$. Starting from the uninformative prior $\text{Beta}(1, 1)$, α_k accumulates positive evidence and β_k accumulates weak evidence. In each round, the system derives positive evidence r_k^{good} and negative evidence r_k^{bad} from the current content score $ContentScore_k$, and updates the posterior as follows:

$$r_k^{good} = \text{clip}\left(\frac{ContentScore_k - \mu_{content}}{\sigma_{content} + \varepsilon}, 0, 1\right), \quad (11)$$

$$r_k^{bad} = 1 - r_k^{good}, \quad (12)$$

$$\alpha_k^{(t)} = \lambda_h \cdot \alpha_k^{(t-1)} + r_k^{good}, \quad (13)$$

$$\beta_k^{(t)} = \lambda_h \cdot \beta_k^{(t-1)} + r_k^{bad}. \quad (14)$$

Here, $\lambda_h \in (0, 1]$ is a historical decay factor. It allows the reputation estimate to adapt to edge-side drift and limits the ability of attackers to build an overly strong reputation through long benign warm-up periods. The long-term utility score is the ex-

pectation of the posterior distribution:

$$HistPerf_k^{(t)} = \frac{\alpha_k^{(t)}}{\alpha_k^{(t)} + \beta_k^{(t)} + \varepsilon} \quad (15)$$

A low *HistPerf* indicates that the node has not recently provided enough positive evidence. The system therefore applies temporary soft isolation, suspending or reducing its aggregation weight for the current round without permanently removing it. This gives legitimate long-tail nodes a chance to recover in later rounds.

(2) Short-Term Risk Stream (Security Veto) Unlike the smoother utility stream, the risk stream responds to sudden events that may indicate an attack. It combines multidimensional security probes, including TEE attestation reports, update magnitude, probe-set loss changes, and trigger activation signals (e.g., r_{probe} , r_{grad} , $r_{trigger}$, r_{pixel}). The strongest probe response in each round is used as the single-round risk value:

$$Risk_k^{inst} = \max\{r_{report}, r_{grad}, r_{probe}, r_{pixel}, r_{trigger}, r_{sign}, r_{peer}, \dots\}, \quad (16)$$

An exponential moving average (EMA) then preserves short-term risk evidence, with $\beta_r = 0.85$:

$$RiskEMA_k^{(t)} = \beta_r \cdot RiskEMA_k^{(t-1)} + (1 - \beta_r) \cdot Risk_k^{inst}. \quad (17)$$

The two streams differ in how they respond to evidence. *HistPerf* smooths short-term fluctuations over multiple rounds, so low similarity caused by long-tail data is not immediately treated as malicious. *RiskEMA*, in contrast, keeps a short memory of abnormal peaks, so a strong risk signal is not immediately diluted by a high historical reputation. The streams meet in *RawScore*: the utility stream controls whether the client should retain a contribution channel, while the risk stream controls whether the current update should be demoted or isolated.

Consider a sleeper-burst poisoning attack. A malicious client may submit normal updates in early rounds and raise *HistPerf_k* close to 1.0. At round T , the same client injects a malicious update with semantic triggers. A single-dimensional score could mask this sudden change because of the client's strong history. In the proposed dual-stream model, however, probes such as $r_{trigger}$ increase *RiskEMA_k* and can push the client across the isolation threshold, blocking the backdoor update before aggregation.

Driven by the two streams, client states move among four states, as shown in Figure 4:

- **Normal Node (NORMAL):** Participates steadily in aggregation. Should risk escalate or historical contribution diminish, the node may be demoted to SUSPECT or QUARANTINE.
- **Suspect Node (SUSPECT):** Placed under closer monitoring. If risk indicators decline, the node may return to NORMAL; if the anomaly persists, stronger isolation is applied.
- **Quarantined Node (QUARANTINE):** Model aggregation privileges are temporarily suspended. After an observation period, the node may recover or move to a more restricted state.

- **Blacklist (BLACKLIST):** Once the hard threshold of the risk stream is triggered, the node is permanently banned, and all subsequent connection requests are terminated at the underlying TEE hardware authentication tier.

The server then combines hardware-level trust (*TrustScore*), current-round update quality (*ContentScore*), and long-term utility (*HistPerf*) into a base score. The short-term risk discount is applied on top of this score to obtain the final aggregation privilege *RawScore_k*:

$$RawScore_k^{base} = (TrustScore_k)^\alpha (ContentScore_k)^\beta \times (HistPerf_k^{(t-1)})^\gamma. \quad (18)$$

$$RawScore_k = RawScore_k^{base} (1 - RiskEMA_k^{(t-1)})^p. \quad (19)$$

where α, β, γ, p control the influence of each dimension. *TrustScore* reflects execution credibility, *ContentScore* reflects current update quality, *HistPerf* captures long-term contribution, and *RiskEMA* acts as a security discount. This fusion prevents low-contribution clients from being permanently removed after a few weak rounds, while still allowing high-risk clients to be demoted or isolated even when their historical utility is high.

State-update summary. The server updates the Beta posterior of *HistPerf_k* using normalized content evidence, updates *RiskEMA_k* from the maximum multidimensional risk probe, and fuses *TrustScore_k*, *ContentScore_k*, *HistPerf_k*, and *RiskEMA_k* into *RawScore_k* for subsequent aggregation control.

4.4. Risk-Gated Layered Aggregation

Stealthy backdoors often affect deep network parameters. A uniform aggregation rule over the whole model can therefore allow unsafe updates to enter sensitive layers. To address this risk, this paper uses layer-wise aggregation with multidimensional risk gating and constrained weight normalization (see Figure 5). Given *RawScore_k* from the previous phase, this stage determines, for each layer, which clients may contribute, how much weight they receive, and whether their update should be clipped.

First, a **Multi-Dimensional Risk Gating Network** estimates the vulnerability of each layer. Let $\mathcal{A}$ represent the set of active clients in the current round that have avoided both soft and hard isolation ($\mathcal{A} = \{k \mid k \notin RiskIsolated \wedge k \notin HistSoftIsolated\}$). For an arbitrary network layer l , its gating sensitivity $S_{total}^{(l)}$ is modeled as a linear combination of privacy, utility, and security terms:

$$S_{privacy}^{(l)} = \exp\left(-\tau_p \frac{l}{L}\right), \quad (20)$$

$$S_{utility}^{(l)} = \frac{\|g_{ref}^{(l)}\|_2}{\max_m \|g_{ref}^{(m)}\|_2 + \varepsilon}, \quad (21)$$

$$S_{security}^{(l)} = 1 - \frac{\sum_{k \in \mathcal{A}} TrustScore_k \cdot \cos(\Delta W_k^{(l)}, g_{ref}^{(l)})}{\sum_{k \in \mathcal{A}} TrustScore_k + \varepsilon}. \quad (22)$$

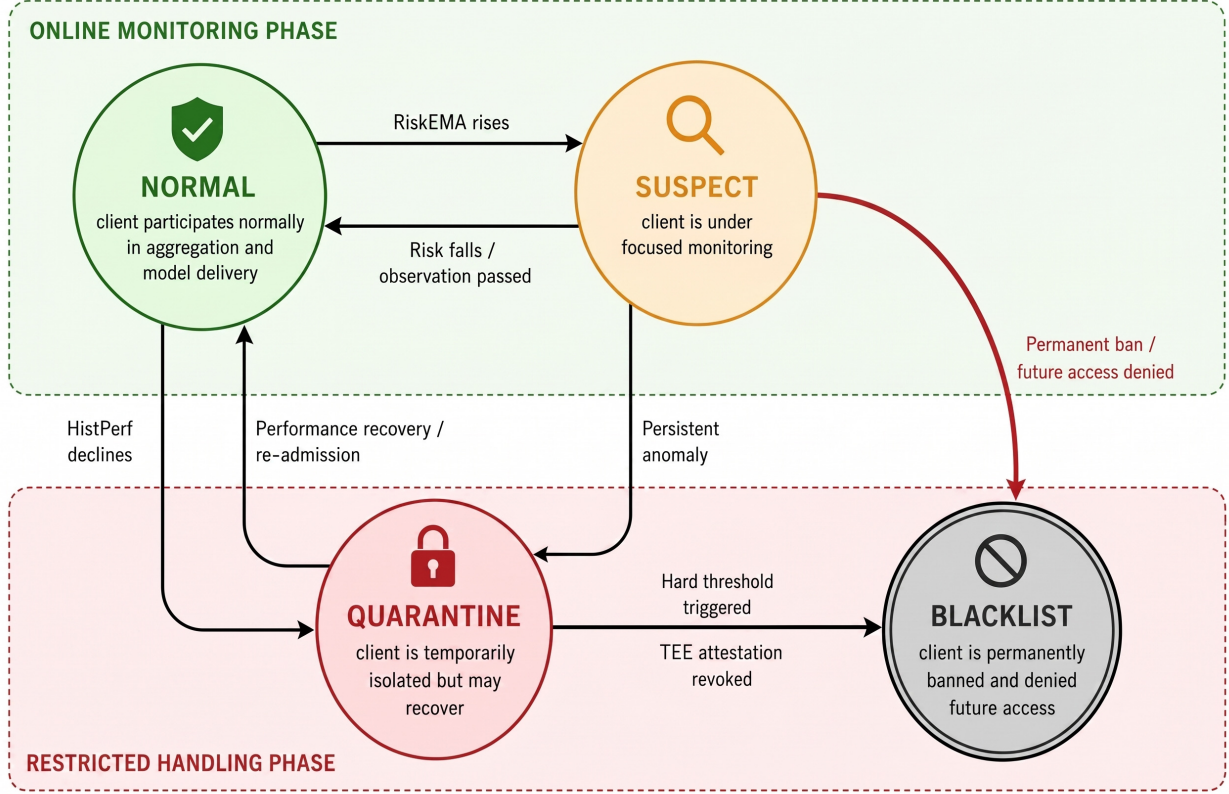


Figure 4: Client supervision state-transition automaton driven by historical utility and short-term risk

The weighted sensitivity is $S_{total}^{(l)} = w_p S_{privacy}^{(l)} + w_u S_{utility}^{(l)} + w_s S_{security}^{(l)}$. Here, $S_{privacy}^{(l)}$ captures layer-level privacy exposure, $S_{utility}^{(l)}$ reflects contribution to the global optimization direction, and $S_{security}^{(l)}$ measures deviation from the reference direction. The gating network uses $S_{total}^{(l)}$ to set the admission threshold $\theta^{(l)} = \mu_{base} + \lambda_s \cdot S_{total}^{(l)}$ and the clipping boundary $C^{(l)} = C_{base} / (S_{total}^{(l)} + \epsilon_c)$. When the risk of a layer increases, $\theta^{(l)}$ rises and $C^{(l)}$ decreases, making it harder for low-trust or high-amplitude updates to affect that layer.

The survivor weights are then re-normalized. After removing high-risk nodes with $RawScore_k < \theta^{(l)}$, the system forms the survivor subset $\Phi^{(l)}$ for the current layer and projects the remaining weights onto a valid aggregation simplex:

$$\tilde{w}_k^{(l)} = \frac{RawScore_k}{\sum_{j \in \Phi^{(l)}} RawScore_j + \epsilon}, \quad \text{s.t.} \quad \sum_{k \in \Phi^{(l)}} \tilde{w}_k^{(l)} \approx 1, \tilde{w}_k^{(l)} \geq 0. \quad (23)$$

This re-normalization step prevents the removal of risky nodes from shrinking the effective layer update.

Finally, the framework applies **Dynamic L2 Clipping and Layered Assembly**. To prevent surviving malicious nodes from pushing the model toward extreme directions, the update of each survivor is projected into a layer-specific L2 ball before

aggregation:

$$\widehat{\Delta W}_k^{(l)} = \frac{\Delta W_k^{(l)}}{\max\left(1, \frac{\|\Delta W_k^{(l)}\|_2}{C^{(l)}}\right)}, \quad \Delta W_{global}^{(l)} = \sum_{k \in \Phi^{(l)}} \tilde{w}_k^{(l)} \cdot \widehat{\Delta W}_k^{(l)}. \quad (24)$$

For example, if an attacker tries to embed semantic triggers in deep fully connected layers, the security sensitivity $S_{security}^{(l)}$ increases. The gate then raises $\theta^{(l)}$ to exclude low-*RawScore* clients and reduces $C^{(l)}$ to limit the norm of surviving updates. Shallow feature extraction layers with lower risk can keep more permissive thresholds and clipping radii. This layer-wise control reduces deep backdoor influence while avoiding unnecessary suppression of useful shallow-layer updates.

Aggregation summary. For each layer, the server computes the privacy, utility, and security sensitivities, derives the layer threshold and clipping radius, selects survivors by *RawScore_k*, re-normalizes their weights, clips their updates, and assembles the secure global increment.

4.5. Model Update Distribution and Knowledge Retention

After layered aggregation, the server applies the secure increment to the global model, $W_{global}^{new} = W_{global}^{old} + \Delta W_{global}$, and distributes the updated model together with client states such as *HistPerf*. This closes the loop between model optimization and trust-state evolution.

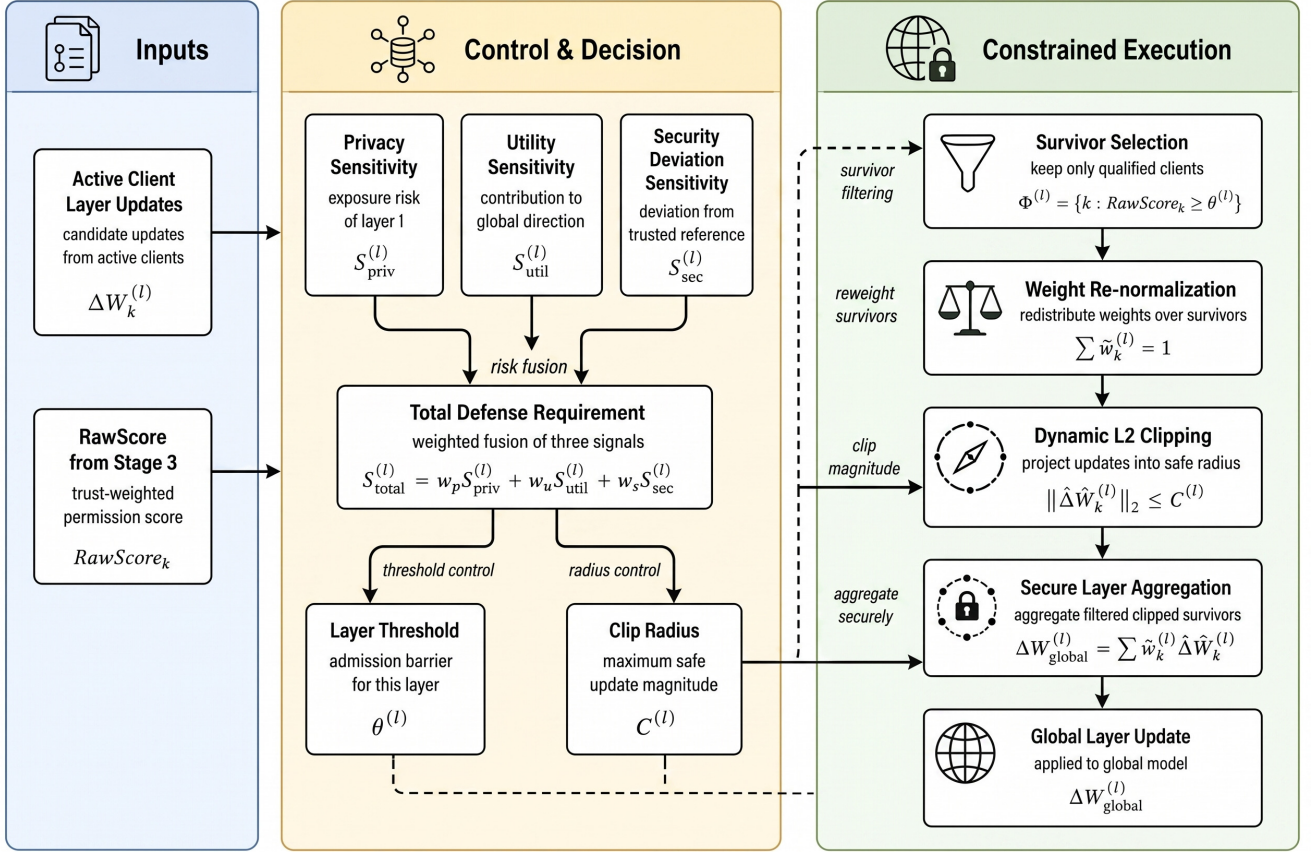


Figure 5: Layered adaptive auditing and dynamic clipping mechanism driven by risk gating

4.6. Theoretical Guarantees of Dual-Stream Orthogonal Decoupling

Traditional single reputation scores, such as accumulated cosine similarity, compress high-dimensional behavior into one axis. Under strong Non-IID data, this worsens the tradeoff between false negative rate (FNR) and false positive rate (FPR). This section analyzes how the dual-stream mechanism reduces this conflict.

Lemma 1 (Variance Convergence Property of Long-Tail Nodes) Assume that the single-round content score $ContentScore_k^{(t)}$ of a legitimate long-tail client k follows a perturbed distribution with expectation μ_{clean} and variance σ_{clean}^2 . Under a rigid threshold τ_{hard} , if $\tau_{hard} > \mu_{clean} - \sigma_{clean}$, this node faces a high probability of sustained elimination. Upon adopting the historical utility stream for smoothing, its steady-state expectation satisfies $\mathbb{E}[HistPerf_k^{(\infty)}] = \mu_{clean}$, and the variance converges to:

$$\text{Var}(HistPerf_k^{(\infty)}) = \frac{1 - \beta_h}{1 + \beta_h} \sigma_{clean}^2 \quad (25)$$

Given $0 < \beta_h < 1$, short-term perturbations are smoothed as $\beta_h \rightarrow 1$. As long as μ_{clean} exceeds the survival threshold, the historical utility stream acts as a low-pass filter, increasing the long-term survival probability of legitimate nodes and supporting $\lim_{t \rightarrow \infty} \text{FPR} = 0$.

Lemma 2 (Short-Term Response to Sleeper Poisoning)

Consider a backdoor node m that uses long benign warm-up rounds followed by intermittent attack bursts. Even if it accumulates high $HistPerf$ early on, an attack round triggers an abnormal probe peak ($\exists r \in \{r_{grad}, r_{probe}, r_{trigger}\}, r \gg 1$). Because $Risk_k^{inst}$ takes the maximum over multiple probes and the EMA is monotone in its input, the RiskEMA trajectory satisfies the following lower bound:

$$RiskEMA_m^{(t)} \geq \max(\beta_r \cdot RiskEMA_m^{(t-1)}, \mathcal{F}_{probe}(r_{grad}, r_{probe}, r_{trigger})). \quad (26)$$

When $\mathcal{F}_{probe}$ detects an anomalous change, the risk value rises quickly and can reach the isolation threshold within a few rounds.

Theorem 1 (Dual-Stream Circuit Breaking and Orthogonal Decoupling) Under the dual-stream mechanism, high-variance updates from honest long-tail nodes are smoothed by the $HistPerf$ channel, while abrupt attack behavior is preserved by the $Risk^{inst}$ channel and can trigger isolation. Processing these signals separately reduces the conflict faced by single-score mechanisms when optimizing both FPR and ASR.

5. Experiments and Analysis

5.1. Experimental Setup and Reproducible Parameters

5.1.1. Baseline Environment and Hardware-Software Stack

Experiments are implemented with Flower 1.5.0 and PyTorch on CIFAR-10 using a lightweight CNN. All simulations run on an Inspur server equipped with dual CPUs and five NVIDIA A10 (24GB) GPUs. The software stack consists of Ubuntu 20.04.6 LTS and CUDA 12.8.

To support reproducibility, each formal run executes 30 communication rounds with 3 local epochs, and complete logs, source codes, and startup scripts are retained. Overall metrics, including Accuracy (Acc) and Attack Success Rate (ASR), are averaged over five independent seeds. Node-state trajectory charts, such as Figure 7, are reported from a representative fixed-seed execution.

5.1.2. Dataset Heterogeneous Partitioning

The 50,000 CIFAR-10 training images are partitioned across 20 admitted clients under a deliberately heterogeneous topology:

- **Shared Base Pool:** A 5,000-sample class-balanced subset is shared among legitimate nodes to provide basic alignment.
- **Weakly Heterogeneous Group (Nodes 6–11):** Client distributions follow Dirichlet $\alpha = 1.0$, simulating moderate distributional skew.
- **Strong Long-Tail Group (Nodes 12–19):** Client distributions follow Dirichlet $\alpha = 0.1$, producing sharply skewed local class profiles and increasing the risk of false rejection by cosine-based defenses.

5.1.3. Malicious Attack Instances and Proportions

To evaluate identification under mixed threats, six malicious nodes are injected into the 20-client training pool, corresponding to a 30% malicious ratio. The attack set covers both data poisoning and parameter manipulation:

- **Client 0 – Label Flip ($\gamma = 0.5$):** Maps a proportion of sample labels to incorrect classes.
- **Client 1 – Trigger Backdoor ($\gamma = 0.2$):** Embeds a trigger into 20% of samples and assigns them to class 0.
- **Client 2 – Clean-Label Backdoor ($\gamma = 0.5$):** Perturbs target-class inputs while preserving their original labels.
- **Client 3 – Semantic Backdoor ($\gamma = 0.5$):** Exploits natural semantic bias to induce spurious associations.
- **Client 4 – Sign-Flipping:** Reverses the returned gradients, i.e., $\Delta \widehat{W}_k = -\gamma \cdot \Delta W_k$.
- **Client 5 – Gradient Scaling:** Amplifies malicious increments by a factor of 100, i.e., $\Delta \widehat{W}_k = 100 \times \Delta W_k$.

Five additional Sybil nodes without valid TEE certificates and three forged-workload free-riders are injected only to validate Phase-One hardware gating and resource-audit screening. These external infiltration or resource-fraud nodes are intercepted before the 20-client statistical analysis: they either fail

TMAA attestation ($M_{attest,k} = 0$) or drive $TrustScore_k \rightarrow 0$ through anomalous workload evidence.

The global hyperparameters and defensive component configurations employed in the experiment are delineated in Table 2.

5.2. Empirical Analysis of Defense Effectiveness

5.2.1. Global Convergence and Attack Prevention Performance

As illustrated in Figure 6, under a 30% malicious proportion, the model accuracy converges steadily to 92.31%, while the backdoor ASR plummets to 10.21%.

Statistical reliability: The aggregate metrics in this section, including Acc and ASR, are reported as the mean of five repeated experiments. Node trajectories are shown from one representative run for readability.

50% malicious-proportion boundary test: To evaluate behavior near the assumed upper bound of malicious participation ($< 50\%$) in Section 3.2, we construct an auxiliary *Flwr-half* setting with a 1:1 adversarial split: 10 malicious and 10 legitimate nodes, covering all six attack categories. The results are shown in Table 3.

Under the boundary setting, the framework maintains 91.81% accuracy and 10.46% ASR, identifies all 10 malicious nodes, and produces no permanent false positives. The 30-round log also shows test loss decreasing from 0.06362 to 0.00883, confirming stable convergence under stringent defensive constraints. The small degradation from the 30% setting is mainly reflected in clean accuracy rather than ASR, which remains close to the random-guess level for CIFAR-10. This pattern indicates that the proposed risk stream does not merely lower ASR by freezing learning; instead, it continues to pass enough benign evidence through the aggregation channel for the global model to converge.

5.2.2. Trust Flow Tracing and Malicious Node Banning Analysis

Trust Flow is intended to expose malicious nodes quickly while managing risk through recoverable intermediate states. To keep the trajectory plot readable, Figure 7 shows four representative nodes: Client 4 (early-stage parameter tampering), Client 0 (high-frequency explicit poisoning), Client 2 (concealed sleeper backdoor), and the legitimate long-tail Client 15. Their state transitions are summarized as follows:

- **Client 0 (Label Flip) & Client 1 (Backdoor):** Both exhibit sustained high risk on r_{grad} and r_{probe} , ultimately triggering `risk_ema_above_0.90_for_4_rounds` at round 8, thereby entering the blacklist.
- **Client 3 (Semantic):** Although the pixel-level perturbation is weak, its update repeatedly deviates from the clean reference g_{root_clean} . It first enters soft isolation and then triggers `risk_soft_isolation_for_8_rounds`, leading to removal at round 16.
- **Client 2 (Clean Label):** Although its local loss appears normal, its persistent drift is captured by the combination rule (`c2_drift_combo_for_5_rounds`), leading to identification and removal by round 24.

Table 2: Global Hyperparameters and Defense Component Configuration

Module	Hyperparameter Description	Value
Base Environment	Total nodes in heterogeneous topology (K)	20
	Local SGD parameters (lr , $Momentum$)	0.001, 0.9
Dual-Stream Phases 1–3	TMAA penalty steepness λ , dead zone τ , decay ρ	5.0, 0.1, 2.0
	2D audit mapping α_1, α_2 and β_{fusion}	0.6, 0.4, 2.0
	History smoothing β_h and risk half-life β_r	0.9, 0.85
Layer Clipping Phase 4	Score fusion exponents (α, β, γ, p)	3.0, 1.0, 0.5, 1.0
	Threshold lower bound μ_{base} and penalty scale λ_s	0.4, 1.2

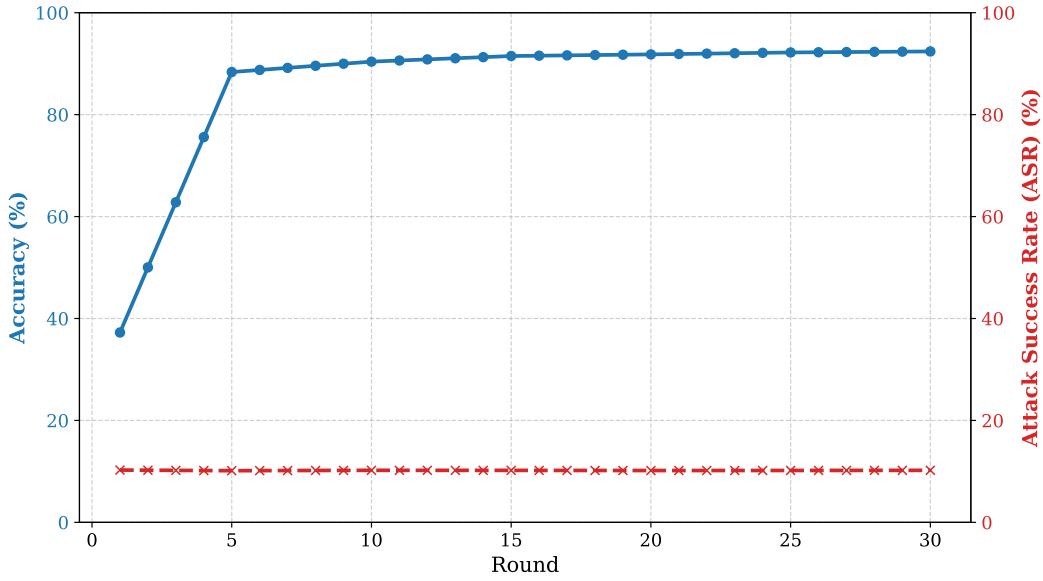


Figure 6: Global Convergence Accuracy and Backdoor ASR Evolution Curves

- **Client 4 (Sign-Flipping) & Client 5 (Gradient Scaling):** Sign flipping sharply reduces $S_{contrib}$ through directional inversion and triggers a hard ban. Gradient scaling is detected through abnormal amplitude and constrained by layer-wise L2 gating before removal.

These trajectories show that the dual-stream mechanism separates low scores caused by data heterogeneity from actual attack behavior. With six malicious nodes, the system achieves a **TPR=100%**. The delayed removal of Client 2 is also useful diagnostically: clean-label poisoning is not always exposed by a single large update, but its repeated small drifts accumulate in the risk probes while the historical utility stream prevents unrelated long-tail clients from being removed by the same rule.

5.2.3. False Positive Protection for Long-Tail Nodes (FPR Analysis)

Traditional distance-centric methods, such as Krum[9] and FLTrust[7], are prone to misjudging legitimate long-tail nodes under strong heterogeneity with Dirichlet $\alpha = 0.1$. In our experiments, **no legitimate node is permanently banned across the 30 training rounds, yielding an FPR of 0% for permanent bans**. Some legitimate nodes, such as Client 15, receive a low

ContentScore early because of data skew and may temporarily enter SUSPECT or QUARANTINE. However, their *RiskEMA* does not continue to rise. They therefore face only temporary admission restrictions and layer-wise clipping. As later contributions improve, *HistPerf* recovers, so legitimate nodes remain in the system.

To further examine this low-FPR behavior, we extract node features before and after intervention and visualize them with t-SNE. Figure 8 shows that the dual-stream mechanism separates malicious clusters from legitimate long-tail clusters while keeping heterogeneous benign nodes connected to the main aggregation region.

5.3. Lateral Baseline Comparison of Defense Mechanisms

Fairness statement: All baselines use the same federated configuration: 20 nodes, Dirichlet $\alpha = 0.1$ data heterogeneity, 30% mixed attacks, the same initialization, optimizer settings ($lr = 0.001$, $momentum = 0.9$), and the same number of communication rounds.

Under this unified setting, we compare the proposed method with four representative robust aggregation techniques:

Table 3: Extreme 50% Malicious Ratio Stress Test vs. Standard 30% Baseline

Setting	Acc (%)	ASR (%)	TPR (%)	FPR (%)
Standard Baseline (30% malicious, 6/20)	92.31 $\pm$ 0.09	10.21 $\pm$ 0.05	100	0.00
Stress Test (50% malicious, 10/20)	91.81 $\pm$ 0.10	10.46 $\pm$ 0.06	100	0.00

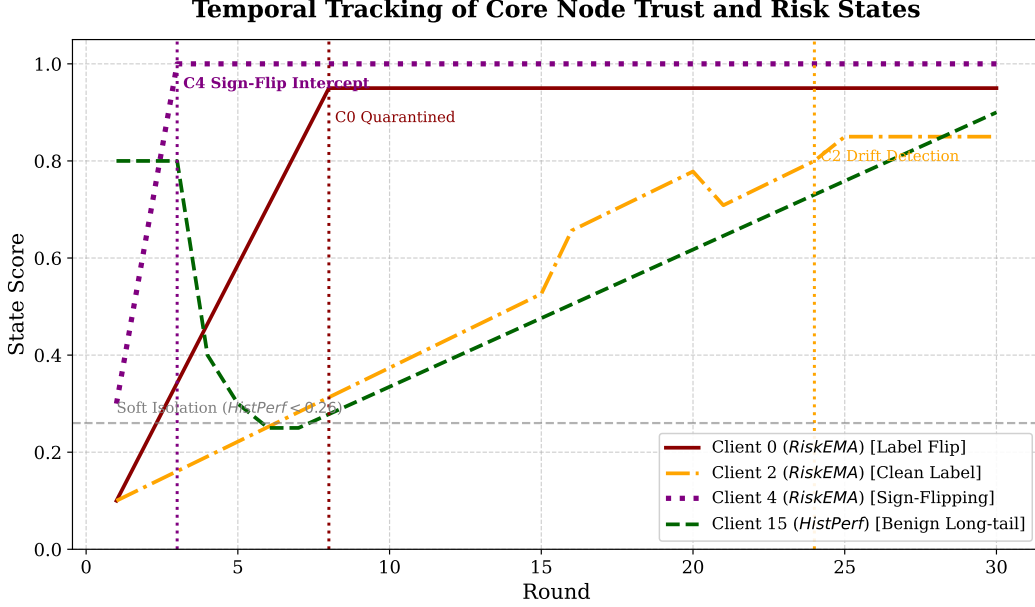


Figure 7: Temporal Tracking of Core Node Trust and Risk States under Mixed Attacks

FedAvg[27] serves as the no-defense baseline and shows how much the model degrades under attack; **Krum**[9] selects one representative update using Euclidean distance; **Trimmed Mean**[10] applies coordinate-wise truncation; and **FLTrust**[7] builds a server-side clean reference vector and weights updates by cosine similarity.

Table 4 reports the quantitative results under this high-risk setting.

FedAvg suffers 92.34% ASR, while Krum suppresses ASR but removes many benign long-tail clients, reducing accuracy to 64.20%. The proposed method reaches 92.31% accuracy, 10.21% ASR, and 0% permanent FPR. A Student’s t -test against strong baselines such as FLTrust indicates statistical significance ($p < 0.05$). The comparison also shows different failure modes. Distance-based aggregation can be effective against large directional outliers, yet it becomes overly conservative when honest updates are dispersed by Dirichlet $\alpha = 0.1$ partitioning. A single clean-reference anchor, as used by FLTrust, improves directional screening but still leaves limited room to distinguish benign long-tail drift from persistent backdoor behavior. Trust Flow adds temporal evidence and layer-wise eligibility control on top of content auditing, which explains why it preserves clean accuracy while keeping ASR low.

5.4. Component-Level Ablation Study

Component ablations examine how dual-stream decoupling, multidimensional probes, and layer-wise re-normalization contribute to robustness and availability. Rather than treating the framework as a black-box combination of defenses, the ablations isolate the role of temporal state modeling, probe coverage, and aggregation recovery after risky updates are removed.

5.4.1. Effectiveness of Dual-Stream Reputation Decoupling

To verify that the system’s low false positive rate (FPR) does not simply come from conservative defensive thresholds, but from the orthogonal decoupling of historical utility and short-term risk, two degraded variants are constructed for comparison: a Single-Stream Exponential Moving Average (Single-Stream EMA) and a HistPerf-Only model. Table 5 reports their interception performance under severe Non-IID conditions and a 30% mixed attack scenario.

Single-stream scoring exposes the threshold dilemma: aggressive settings raise FPR, while conservative settings miss sleeper attacks. HistPerf-only protects long-tail clients but lacks short-term risk sensitivity. Dual-stream decoupling achieves 98.50% TPR with only 0.05% FPR. These results support the design choice of keeping *HistPerf* and *RiskEMA* independent until the final privilege score is computed. A benign long-tail client can survive because weak contribution evidence

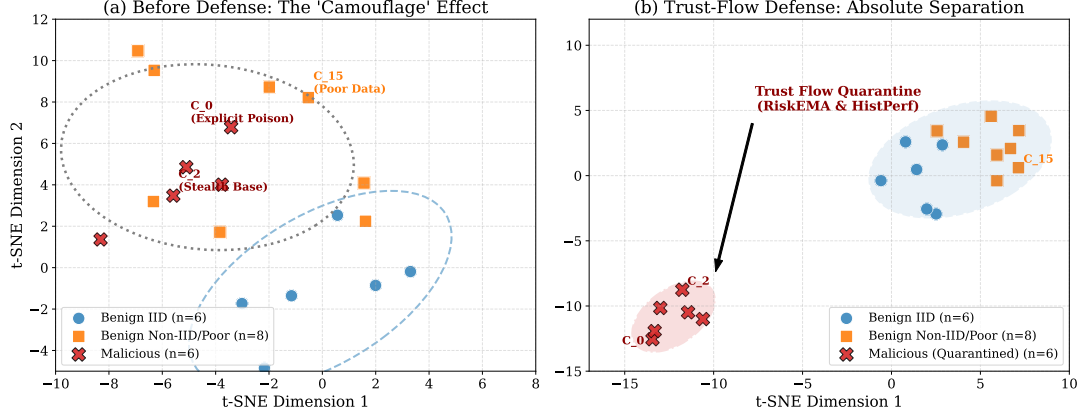


Figure 8: t-SNE Visualization of Node Feature Vectors Before and After Dual-Stream Defense

Table 4: Performance Comparison of FL Defense Strategies under 30% Mixed Attacks

Defense Strategy	Core Mechanism	Global Acc. $\mu \pm \sigma$ (Acc. %)	ASR $\mu \pm \sigma$ (ASR %)	Benign FPR μ (FPR %)
FedAvg [27]	No defense baseline	86.15 ± 1.2	92.34 ± 5.1	N/A (*)
Krum [9]	Euclidean distance	64.20 ± 2.8	12.15 ± 0.9	64.20 (**)
Trimmed Mean [10]	Coordinate-wise trimming	71.35 ± 2.4	38.60 ± 4.2	N/A (*)
FLTrust [7]	Unidirectional trust anchor	81.25 ± 1.5	15.42 ± 1.1	21.40 (**)
Proposed (Trust Flow)	Dual-stream + layered clipping	92.31 ± 0.09	10.21 ± 0.05	0.00

(*) Note: FedAvg and Trimmed Mean do not include any explicit client trust evaluation or permanent blacklist removal (Ban) mechanism. Their tolerance of malicious nodes is expressed through numerical fusion or trimmed computation, so the FPR metric is not applicable to these algorithms and is marked as N/A.

(**) Note: The false positive rate (FPR) for benign nodes is strongly constrained by deterministic rules and the fixed heterogeneity of data samples under the Dirichlet distribution. With a fixed dataset partition, the defense rules are deterministic, and the variance observed for this type of mistaken removal across repeated experiments is extremely small. This table therefore reports only the stably converged scalar expected mean μ for this metric. The value 0.00 here specifically refers to permanent false positives. Mild isolation for nodes temporarily placed in the suspicious pool is not counted as an irreversible permanent failure of the model because it allows recovery after the score rebounds.

is smoothed across rounds, whereas a sleeper attacker cannot hide a sudden probe peak behind earlier high utility.

5.4.2. Multi-Dimensional Risk Probe Ablation

Figure 9 contrasts the ASR suppression capabilities of Vanilla Clean-Root, shallow statistical probes, and full-dimensional deep probes, and also reports the convergence benefit of survivor weight re-normalization.

Shallow probes handle sign flipping and scaling, but clean-label backdoors remain difficult because their gradients are nearly collinear with benign updates. Adding cross-entropy and high-level activation probes reduces clean-label ASR below 9.2%. The gain comes from using complementary evidence: r_{grad} captures geometric and norm anomalies, r_{probe} measures whether a tentative update harms clean probe loss, and $r_{trigger}$ detects latent feature shifts that may not appear as a large cosine deviation. The max-fusion rule keeps the strongest warning signal, so a concealed backdoor can still be stopped even when most surface-level statistics look benign.

5.4.3. Layered Gating and Re-Normalization Convergence Benefits

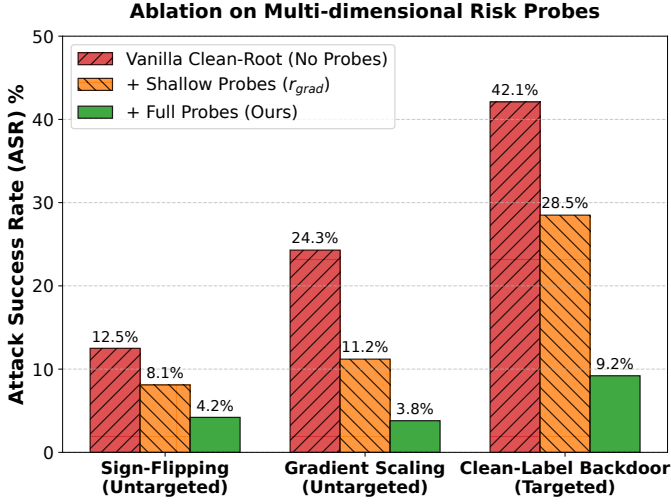
Global clipping suppresses useful shallow-layer updates and stalls near 85% accuracy. Layer-wise gating improves accuracy to about 89%, but without weight re-normalization the survivor weights sum below one, creating implicit learning-rate decay. Trust Flow re-normalizes survivors and recovers accuracy close to the attack-free level. This behavior is important for edge FL because the model must keep learning after malicious clients are isolated. If removal only reduces the update magnitude, the defense may appear secure while silently slowing convergence; the re-normalization step prevents this loss of optimization momentum.

5.5. Sensitivity Analysis on Data Heterogeneity

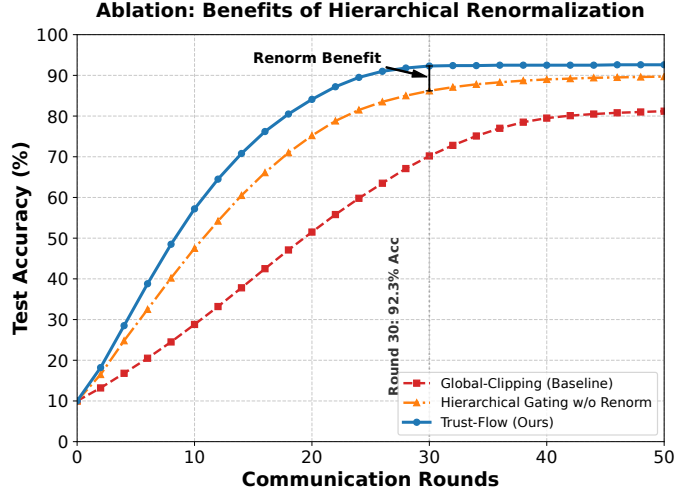
Many robust aggregation methods work well when client data are close to IID, but degrade as heterogeneity increases. To evaluate this effect, we test global accuracy under four Dirichlet settings ($\alpha = 100, 1.0, \dots, 0.1$), as shown in Figure 10.

Table 5: Ablation Analysis of Orthogonal Decoupling on FPR and Recall

Reputation Architecture	Threshold Control	FPR	TPR
Single-Stream EMA (Baseline)	Aggressive (anti-poison)	18.25%	95.12%
	Conservative (anti-FP)	2.10%	48.33%
HistPerf-Only	Dynamic adaptive	0.15%	21.05%
Dual-Stream Decoupling (Proposed)	Dynamic adaptive	0.05%	98.50%

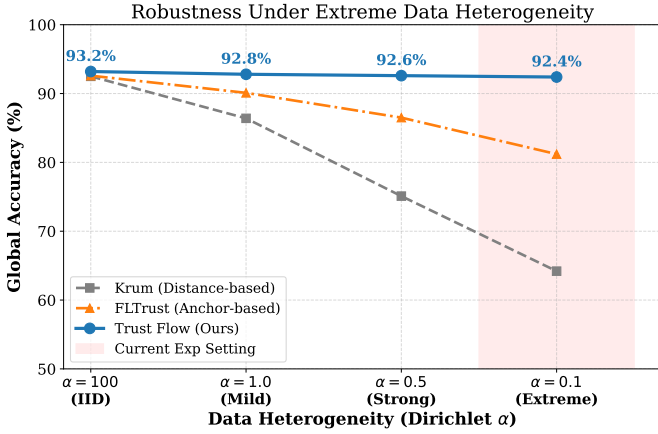


(a) Multi-dimensional risk-probe ablation.



(b) Weight re-normalization effect.

Figure 9: Component-level ablation of risk probes and re-normalization.

Figure 10: Robustness Stress Test under Varying Data Heterogeneity (Dirichlet α)

The curves show that when heterogeneity is weak ($\alpha = 100$ or $\alpha = 1.0$), Krum and FLTrust perform close to the proposed method, with Acc > 90%. Under the extreme long-tail setting ($\alpha \rightarrow 0.1$), however, distance-based filtering degrades sharply, with Krum falling to 64.2% accuracy. The proposed method remains close to 92% accuracy under the same setting, indicating that the dual-stream design is more robust to severe heterogeneity.

We also test the fusion parameter β_{fusion} at 0.5, 1.0, 2.0, and 3.0. A smaller value keeps more long-tail clients but increases

ASR, whereas a larger value suppresses attacks more strongly but can over-penalize heterogeneous benign updates. Based on the Acc/ASR tradeoff, we set $\beta_{fusion} = 2.0$. This value gives the clean reference enough influence to counter collusive drift, while still allowing peer consistency and historical utility to protect legitimate clients whose local class distribution is far from the global average.

5.6. System Computation and Communication Overhead Analysis

Although the framework uses a multi-phase evaluation protocol, its computation and communication overhead remains modest. Table 6 shows that the edge side only generates a TEE quote and performs lightweight monitoring, adding roughly 12 ~ 15 ms. Each upload appends a TrustReport of about 15 KB. Compared with model transfers of approximately ~ 10 MB, this adds less than 0.15% bandwidth overhead.

Table 6: Edge-Cloud Overhead: Standard FL vs. Proposed Framework (20 Clients)

Metric	FedAvg	Proposed Framework
Edge extra latency	0 ms	~ 15 ms (TEE signing only)
Upload extra payload	0 KB	~ 15 KB (appended report)
Cloud aggregation time	~ 2.10 s	~ 4.85 s (audit & dual-stream)

The main extra cost lies on the server: aggregation increases from 2.10 s to 4.85 s per round. Since WAN-based FL rounds

usually spend tens of seconds on communication and synchronization, this overhead remains practical. The design also keeps edge-side requirements limited: clients only need to provide attestation evidence and compact runtime metadata, while the heavier consistency checks, probe evaluation, EMA updates, and layer-wise filtering are placed on the cloud side. This split is consistent with heterogeneous edge deployments, where client devices differ widely in compute budget.

6. Conclusion and Future Work

This paper presented Trust-Flow TFL, a framework for securing edge federated learning under severe Non-IID data and compound attacks. The framework combines TEE-anchored admission, dual-stream reputation evolution, and layer-wise robust aggregation. By modeling historical utility and short-term risk separately, it reduces false positives for benign long-tail clients while maintaining high attack recall.

Under the 30% mixed-attack setting, the framework achieves 92.31%(±0.09%) accuracy and limits ASR to 10.21%(±0.05%). In the 50% boundary stress test, it maintains 91.81% accuracy and 10.46% ASR, with 100% TPR and 0% FPR for permanent bans. These results show that the method remains effective under strongly adversarial and heterogeneous conditions.

Limitations and future work: Because several emerging TEE-FL methods do not provide publicly verifiable implementations, this study does not include them as direct baselines under the same experimental setting. The threat model also follows standard TEE assumptions and does not cover microarchitectural side channels or physical probing. We plan to release the source code and experimental scripts with the final manuscript to support reproduction. Future work will extend the multidimensional risk probes beyond computer vision to NLP and federated fine-tuning scenarios, evaluate their generalization across modalities, and examine larger client populations with more diverse device capabilities.

Appendix A. Detailed Computation of Short-Term Risk Probes

To support reproducibility and transparency, this appendix gives the calculation details of the core probes (r_{grad} , r_{probe} , $r_{trigger}$) used by the short-term risk stream of the proposed Trust-Flow framework. During the t -th aggregation round, the probes are defined as follows.

Appendix A.1. Gradient Direction and Amplitude Probe (r_{grad})

The r_{grad} probe quantifies the geometric deviation of the client update $\Delta W_k^{(t)}$ relative to the reference direction $g_{root}^{(t)}$, with a particular focus on sign flipping and amplitude anomalies.

This metric evaluates both directional deflection and scale mutation:

$$r_{grad,k}^{(t)} = \lambda_d \left(1 - \frac{\Delta W_k^{(t)} \cdot g_{root}^{(t)}}{\|\Delta W_k^{(t)}\| \cdot \|g_{root}^{(t)}\|} \right) + \lambda_m \max \left(0, \frac{\|\Delta W_k^{(t)}\|_2 - \mu^{(t)}}{\sigma^{(t)}} \right). \quad (\text{A.1})$$

Here, λ_d and λ_m control the directional and amplitude terms, respectively, and are set to 0.5 by default. The terms $\mu^{(t)}$ and $\sigma^{(t)}$ denote the mean and standard deviation of gradient L2 norms across admitted clients in round t .

Appendix A.2. Small-Sample Cross-Entropy Probe (r_{probe})

Distance statistics alone are insufficient for identifying targeted poisoning against minority classes. The r_{probe} metric uses a small clean probe set on the server, denoted by $\mathcal{D}_{probe}$, and measures the cross-entropy loss surge before and after a client update is tentatively applied:

$$r_{probe,k}^{(t)} = \text{ReLU} \left(\mathcal{L}_{CE}(W_{global}^{(t-1)} + \Delta W_k^{(t)}; \mathcal{D}_{probe}) - \mathcal{L}_{CE}(W_{global}^{(t-1)}; \mathcal{D}_{probe}) \right). \quad (\text{A.2})$$

Appendix A.3. Deep Neuron Backdoor Activation Probe ($r_{trigger}$)

For concealed threats such as clean-label attacks, conventional statistical indicators and validation losses may remain deceptively stable. The $r_{trigger}$ probe therefore examines the distributional shift of high-level feature activations. Let A_k denote the pre-activation outputs of the client model on the validation set, and let $\mathcal{H}(\cdot)$ denote Softmax normalization into a probability simplex. The trigger risk is computed as:

$$r_{trigger,k}^{(t)} = \text{KL}(\mathcal{H}(A_{base}) \parallel \mathcal{H}(A_k)). \quad (\text{A.3})$$

After normalization and fusion, these anomalous signals form the single-round risk extremum $Risk_k^{(t)}$, which enters the EMA pipeline and supports fast circuit breaking in subsequent trust-state decisions.

Appendix A.4. Probe Normalization and Circuit-Break Fusion

The three probes above are deliberately heterogeneous: r_{grad} focuses on geometric deviation, r_{probe} reflects functional loss degradation, and $r_{trigger}$ captures latent activation drift. Directly averaging these signals would weaken sharp but security-critical outliers. The implementation therefore applies a monotone normalization to each probe and then preserves the strongest abnormal evidence through a max-fusion rule:

$$Risk_k^{inst} = \max\{r_{report}, r_{grad}, r_{probe}, r_{pixel}, r_{trigger}, r_{sign}, r_{peer}\}. \quad (\text{A.4})$$

This design gives the risk stream a veto role. A client with high historical utility may still be demoted when any single probe exposes a sudden backdoor-like mutation, while a benign long-tail client is not permanently removed simply because one round

yields a low contribution score. The resulting single-round risk evidence is further smoothed by EMA,

$$RiskEMA_k^{(t)} = \beta_r RiskEMA_k^{(t-1)} + (1 - \beta_r) Risk_k^{inst}, \quad (A.5)$$

which retains short-term attack evidence across adjacent rounds without converting isolated benign fluctuations into irreversible bans. In practice, this fusion step also limits communication overhead: the edge side only appends compact audit metadata, while the cloud side performs the probe comparison, EMA update, and subsequent privilege adjustment.

Appendix A.5. Operational Coupling with State Transitions

The normalized risk stream is coupled with the client-state machine rather than being used as an isolated scalar filter. During each round, clients with low historical utility but bounded risk are first routed to temporary observation states, where their aggregation privileges are reduced or suspended without immediately terminating their participation. This handling is important for strongly Non-IID edge clients, because a low current-round contribution may reflect class imbalance or long-tail local data rather than poisoning behavior. In contrast, clients that repeatedly trigger high-risk probes are escalated from SUSPECT or QUARANTINE toward permanent isolation, even if their historical utility remains high because of earlier benign behavior.

This coupling also clarifies the division of responsibility between the edge and cloud sides. Edge devices provide trusted execution evidence and compact runtime reports, while the cloud server evaluates cross-client consistency, probe-set loss shifts, activation drift, and layer-wise aggregation eligibility. The final state decision therefore combines physical trust, update quality, temporal utility, and instant security evidence. Such a design avoids two common failure modes in trusted federated learning: over-trusting a device merely because it passes attestation, and over-penalizing legitimate heterogeneous clients merely because their updates are statistically distant from the majority.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRediT authorship contribution statement

Jiakai Zhou: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Visualization, Writing – original draft. Chao Bu: Supervision, Project administration, Funding acquisition, Writing – review & editing.

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Data availability

The data and materials that support the findings of this study are available from the corresponding author upon reasonable request.

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